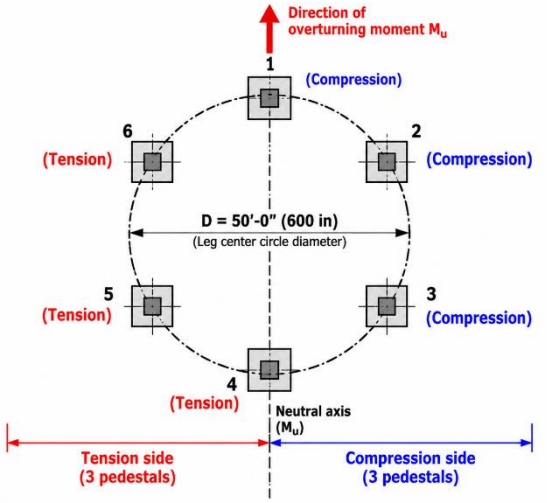
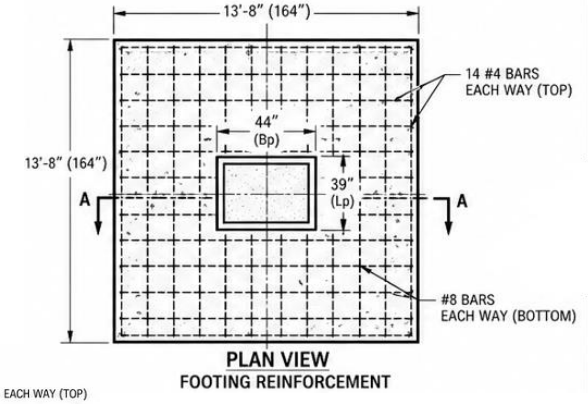
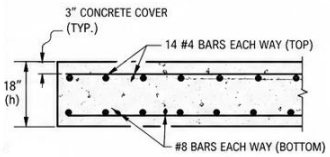
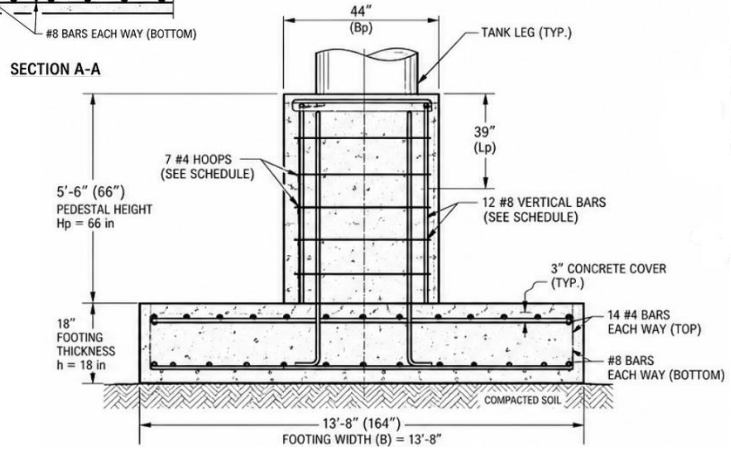
4. Footing segment

No	Design Parameter	Symbol	Value	Source
1	Total factored axial load	P_u	4623.218 kips	Main Tank Design
2	Factored overturning moment	M_u	3823.08 kips-ft	Main Tank Design
3	Factored shear	V_u	62.4134 kips	Main Tank Design
4	Number of legs	N_{leg}	6	Tank Geometry
5	Pedestal size	$B_p \times L_p$	44 × 39 in	Pedestal Design
6	Pedestal height	H_p	5'-6" = 66 in	Pedestal Design
7	Concrete strength	f'_c	4 ksi	User Input
8	Reinforcing steel	f_y	60 ksi	User Input
9	Concrete unit weight	γ_c	150 pcf	From ACI Code
10	Assumed footing thickness	h	18 in	Assumed
11	Concrete cover	c_c	3 in	Assumed
12	Footing dimensions	$B \times L$	13′-8″ × 13′-8″	User Input
13	Top reinforcement		14 #4 bars each way	User Input
14	Pedestal reinforcement		12 #8 vertical bars	User Input
15	Pedestal hoops		7 #4 hoops	User Input
16	Bottom reinforcement		#8 bars, each way	User input
17	Allowable soil pressure (Assumed)	q_{allow}	3.0 ksf	User Input









Step 1: Footing Load

Gravity load per leg

$$P_{\text{leg}} = \frac{P_u}{N_{\text{leg}}}$$

$$P_{leg} = \frac{4623.218}{6}$$
$$P_{leg} = 770.536 \text{ kip}$$

Self-weight of Pedestal

$$W_p = \gamma_c \left(b_p * h_p * L_p\right) = \left(\frac{44}{12}\right)\left(\frac{39}{12}\right)\left(\frac{66}{12}\right)(0.150)$$
$$W_p = 9.831 \text{ kip}$$

Factored self-weight, assuming a 1.2 dead-load factor:

$$W_{p,u} = 1.2(9.831) = 11.797 \text{ kips}$$

$$P_c = P_{leg} + W_p$$
$$P_c = 770.536 + 11.797 = 782.333 \text{ kip}$$

Step 2: Preliminary Footing Size

Required footing area

$$B = L = 13 + \frac{8}{12} = 13.6667 \text{ ft}$$

$$\text{A} = \frac{P}{q_{allow}}$$

$$A_{req} = \frac{782.33}{3} = 260.778 ft^2$$

$$\boldsymbol{A_f = 186.778 \text{ ft}^2}$$

Note: This sizing uses a factored load with an allowable soil pressure, which is not a consistent final design check. Final sizing must use the applicable service-load combination for allowable bearing and include the overturning-moment effect.

Step 3: Soil Bearing Check

Average factored bearing pressure:

$$q_u = \frac{P_u}{A_f}$$

$$q = \frac{782.33}{186.778}$$

$$q = 4.189 \text{ ksf}$$

This is the simplified average pressure using factored load, not the final allowable-soil-pressure check.

Step 4: Footing Thickness

$h = 18$ in., $c_c = 3$ in., $d_b = 1.00$ in.

For #8 bars use Table A-1:

$d_b = 1.0$ in.

Therefore:

$$d = h - Cc - \frac{d_b}{2}$$

$$d = 18 - 3 - \frac{1.0}{2}$$

$$d = 14.5 \text{ in.}$$

Step 5: Two-Way Shear / Punching Shear

[ACI 318-19: 22.6.4.1, 22.6.5.2 & 3]

Pedestal dimensions:

$$c_1 = 44 \text{ in.}, \quad c_2 = 39 \text{ in.}$$

$$b_o = 2(c1 + c2 + 2d)$$

$$b_o = 4(44 + 39 + 29)$$

$$b_o = 224 \text{ in}$$

Area inside the critical perimeter:

$$A_o = (c1 + d)(c2 + d)$$

$$A_o = (44 + 14.5)(39 + 14.5)$$

$$A_o = 3129.75 \text{ in}^2 = 21.7344 \text{ ft}^2$$

Soil reaction inside:

$$P_{soil} = q_u A_o$$

$$P_{soil} = 4.189(21.7344)$$

$$P_{soil} = 91.03 \text{ kip}$$

Concrete exposure	Member	Reinforcement	Specified cover, mm
Cast against and permanently in contact with ground	All	All	75
Exposed to weather or in contact with ground	All	No. 19 through No. 57 bars	50
		No. 16 bar, MW200 or MD200 wire, and smaller	40
Not exposed to weather or in contact with ground	Slabs, joists and walls	No. 43 and No. 57 bars	40
		No. 36 bar and smaller	20
	Beams, columns, pedestals and tension ties	Primary reinforcement, stirrups, ties, spirals and hoops	40

Table 20.6.1.3.1 Specified concrete cover for cast-in-place nonprestressed concrete members

ASTM Standard Reinforcing Bar — Nominal Diameter

Bar designation	Nominal diameter, d_b (in.)
#3	0.375
#4	0.500
#5	0.625
#6	0.750
#7	0.875
#8	**1.000**
#9	1.128
#10	1.270
#11	1.410
#14	1.693
#18	2.257

Source: ACI 318-14, Appendix A—Steel Reinforcement Information; ASTM standard reinforcing bars.

Punching shear demand:

$$V_u = P_{u,\text{leg}} - P_{\text{soil}}$$

$$V_u = 782.33 - 91.03$$

$$V_u = 691.30 \text{ kip}$$

Using f'c=4000 psi, $\lambda = 1.0$, $\phi = 0.75$ and the minimum of the three ACI punching-shear expressions:

$$\phi V_c \approx 616.26 \text{ kips}$$

Table 19.2.4.1(a)—Values of λ for lightweight concrete based on equilibrium density

w_c, lb/ft³	λ	
≤ 100	0.75	(a)
100 < w_c ≤ 135	0.0075w_c ≤ 1.0	(b)
> 135	1.0	(c)

Therefore:

$$D/C = \frac{691.30}{616.26}$$

$$D/C \approx 1.122$$

For a rectangular loaded area, define:

$$\beta = \frac{44}{39} = 1.128$$

location of pedestal

$\alpha_s$depends on where the loaded area is located (ACI 318-19 22.6.5.3):

Pedestal location	α_s
Interior	**40**
Edge	**30**
Corner	**20**

Your footing is being designed as an interior/square pedestal condition, so:

$\alpha_s = 40$

Equation 1: where $\beta \leq 2$

$$\phi V_c = 0.75\left(4\lambda_s\lambda\sqrt{f_c'}\right)b_o d$$

However, for **isolated footings**, the size-effect factor may be taken as:

$$\lambda = 1.0$$

The size-effect factor may be neglected for shallow foundations based ACI-318-19.

Thus:

$$\phi V_c = 0.75(4)(1)(1)\sqrt{4000}(224)(14.5)/1000$$

$$\phi V_{c1} = 616.26\ kips$$

The other two expressions give approximately:

Equation 3

$$\phi V_c = 0.75\left(2 + \frac{\alpha_s d}{b_o}\right)\lambda_s \lambda \sqrt{f'_c} b_o d$$

$$\phi V_{c3} = \frac{0.75\left(2 + \frac{40(14.5)}{224}\right)(1)(1)\sqrt{4000}(224)(14.5)}{1000}$$

$$\phi V_{c3} = 705.7 kip$$

Therefore:

$$\phi V_c = \min(616.26,\ 854.5,\ 705.7)$$

$$\phi V_c = 616.26\ kip$$

Check:

$$D/C = \frac{V_u}{\phi V_c}$$

Utilization:

$$\frac{691.30}{616.26} = 1.122$$

$$Utilization = 112.2\%$$

Note: PUNCHING SHEAR — FAIL. User Can Increase footing width to decrease utilization.

Step 7: One-Way Shear

The footing projection from the pedestal face to the footing edge is:

$$l1 = \frac{B - c}{2} = \frac{13.6667 - 44/12}{2} = 5.00\ ft$$

$$l1 = \frac{B - c}{2} = \frac{13.6667 - 39/12}{2} = 5.2083\ ft$$

$$d = \frac{14.5}{12} = 1.2083\text{ft}$$

The factored soil pressure obtained previously is:

$$q_u = 4.189\ ksf$$

Therefore, for a 1-ft-wide strip:

$$V_{u1} = q_u(l_1 - d)$$
$$V_{u1} = 4.189(5.00 - 1.2083)$$
$$V_{u1} = 15.88\ kip/ft$$
$$V_{u2} = q_u(l_2 - d)$$
$$V_{u2} = 4.189(5.2083 - 1.2083)$$
$$V_{u2} = 16.76\ kips/ft$$

For normal-weight concrete, $\lambda = 1.0$, and:

$$V_c = 2\lambda\sqrt{f_c'}b_w d$$
$$V_c = 2(0.75)\sqrt{4000}(12)(14.5)/1000$$
$$V_c = 16.51\ kip/ft$$

Comparison:

- Direction 1: 15.88<16.51— preliminary pass.
- Direction 2: 16.76>16.51— preliminary fail.

Step 8: Flexure and Reinforcement

Based on the drawing design the same bottom reinforcement as the drawing: 14 #8 bars each way. reinforcement per foot, using approximately 12.08 in. spacing:

$$A_{s,\text{prov}} = 0.79\frac{12}{12.08}$$
$$A_{s,\text{prov}} \approx 0.785\ \text{in}^2/\text{ft}$$

For the simplified uniform-pressure model:

$$M_u = \frac{q{l_1}^2}{2}$$
$$M_{u1} = \frac{4.189(5.000)^2}{2}$$

$$M_{u1} = 52.36\ kip - ft/ft$$

$$M_u = \frac{q{l_2}^2}{2}$$

$$M_{u2} = \frac{4.189(5.2083)^2}{2}$$

$$\boldsymbol{M_{u2} = 56.78 - kip - ft/\text{ft}}$$

Using d=14.5 in., fy=60 ksi, f'c=4 ksi, and $\phi = 0.90$:

$$a = \frac{A_s f_y}{0.85 f_c' b}$$

$$a = \frac{0.785(60)}{0.85(4)(12)}$$

$$\boldsymbol{a \approx 1.154\ \text{in}}$$

$$M_n = A_s f_y \left(d - \frac{a}{2}\right)$$

$$\phi A_s f_y \left[d - \frac{A_s f_y}{2(0.85 f_c' b)}\right] = \phi M_n$$

$$\phi M_n = 0.90(0.785)(60)\left(14.5 - \frac{1.154}{2}\right)$$

$$\phi M_n \approx 590.2\ \text{kip} - \text{in./ft}$$

$$M_{u1} = 52.36(12) = 628.3\ \text{kip} - \text{in./ft}$$

$$M_{u2} = 56.78(12) = 681.4\ \text{kip} - \text{in./ft}$$

Both exceed the calculated flexural capacity.

Flexure: FAIL in both directions under this simplified model.

Step 9: Pedestal-to-footing bearing

$$A_1 = 44(39) = 1716 \text{ in}^2$$

$$\phi B_n = 0.65(0.85)(4)(1716)$$

$$\phi B_n = 3792.36 \text{ kips}$$

$$D/C = \frac{782.333}{3792.36}$$

$$\boldsymbol{D/C = 0.206}$$

Pedestal bearing preliminary pass.

Step 10: Dowel / Pedestal-to-Footing Reinforcement

For a cast-in-place pedestal connected to a foundation, ACI 318-14 $16.3.4.1 states:

$$A_{s,\text{dowel}} \geq 0.005 A_g$$

where A_g is the gross cross-sectional area of the supported pedestal/column and at least four dowels.

Based on pedestal:

$$A_{s,min} = 0.005(1521)$$

$$A_{s,\text{min}} = 0.005(1716) = 8.58 \text{ in}^2$$

Adopted 18#6:

$$A_s = 12(0.79) = 9.48 \text{ in}^2$$

Therefore:

$$9.48 > 8.58 \ldots\ldots \textit{DOWEL/INTERFACE STEEL — PASS}$$

Step 11: Development of deformed bars in compression

$$l_{dc} = \frac{f_y \psi_r}{50\lambda\sqrt{f_c'}} d_b$$

with the minimum requirement:

$$0.0003 f_y d_b \geq 8 \textit{ in.}$$

For #6:

$$d_b = 1.0 \textit{ in.}$$

$$l_{dc} = \frac{60{,}000(1.0)}{50(1)\left(\sqrt{4000}\right)}(1.0)$$

$l_{dc} \approx 18.97$ in.

Minimum:

$$0.0003(60,000)(1.0) = 18.00\ in.$$

Therefore:

$$l_{dc} = 18.00 \text{ in.}$$

For detailing, we can adopt:

$$18.97 > 18$$